DATABASE MANAGEMENT SYSTEMS LABORATORY

PRACTICAL LABORATORY EXAMINATION

Name:	**Register No.:**
Date:	**Set:**

Time: 2 Hours **Maximum Marks: 15**

Instructions:

- Execute all the required SQL commands in the database.
- Display the output wherever required.
- Use appropriate SQL syntax and constraints.
- Save the commands and outputs for verification.

SET 1
LIBRARY DATABASE

Question: Create and manipulate a `BOOK` table using SQL commands.

1. Create the following table:

```
BOOK(
    BOOK_ID NUMBER(5) PRIMARY KEY,
    TITLE VARCHAR2(50),
    AUTHOR VARCHAR2(30),
    PRICE NUMBER(6,2),
    CATEGORY VARCHAR2(20)
)
```

2. Insert **5 records** into the table.
3. Add a column `PUBLISHER VARCHAR2(40)` using the `ALTER` command.
4. Display all the records in the table.
5. Write SQL queries to display:
   (a) Books with price greater than 500.
   (b) Books with price between 300 and 800.
6. Display the books whose title starts with the letter `D`.
7. Display all books in descending order of price.
8. Find the following using aggregate functions:
   (a) Average price of books.
   (b) Minimum price.
9. Delete one book from the table.

— End of Set 1 —

SET 2
EMPLOYEE DATABASE

Question: Create and manipulate an `EMPLOYEE` table using SQL commands.

1. Create the following table with appropriate constraints:

```
EMPLOYEE(
    EMP_ID NUMBER(5) PRIMARY KEY,
    EMP_NAME VARCHAR2(30) NOT NULL,
    EMAIL VARCHAR2(50) UNIQUE,
    SALARY NUMBER(8,2) CHECK(SALARY > 10000),
    CITY VARCHAR2(20) DEFAULT 'KOCHI'
)
```

2. Insert **5 employee records** into the table.
3. Display all employee records.
4. Write SQL queries to display:
   (a) Employees with salary greater than 30000.
   (b) Employees with salary between 20000 and 50000.
5. Display employees whose names start with the letter `A`.
6. Display employees in descending order of salary.
7. Find the following using aggregate functions:
   (a) Total salary.
   (b) Average salary.
8. Update the salary of one employee and delete another employee.
9. Attempt to insert a duplicate `EMP_ID` and observe the constraint violation.

— End of Set 2 —

SET 3
STUDENT RESULT DATABASE

Question: Create a `MARKLIST` table and perform the required operations.

1. Create the following table:

```
MARKLIST(
    REGNO NUMBER(5) PRIMARY KEY,
    NAME VARCHAR2(50),
    PHYSICS NUMBER(3),
    CHEMISTRY NUMBER(3),
    MATHS NUMBER(3),
    TOTAL NUMBER(4)
)
```

2. Insert **5 student records** into the table.
3. Display all records in the table.
4. Display `REGNO`, `NAME`, and `TOTAL`.
5. Write SQL queries to find:
   (a) Students scoring above 90 in Physics.
   (b) Students whose total mark is above 250.
6. Display students whose total mark is between 210 and 240.
7. Find the complete details of a student with a specified name.
8. Update the Maths mark of the selected student and recalculate the total.

— End of Set 3 —

SET 4
COLLEGE STUDENT DATABASE

Question: Create and manipulate a `STUDENT` table.

1. Create the following table:

```
STUDENT(
    STD_ID NUMBER(5) PRIMARY KEY,
    NAME VARCHAR2(50) NOT NULL,
    BRANCH VARCHAR2(10),
    PLACE VARCHAR2(30),
    BATCH VARCHAR2(5)
)
```

2. Insert **5 student records** with different branches, places, and batches.
3. Add a column `PHONE VARCHAR2(15)`.
4. Rename the column `PHONE` to `PHNO`.
5. Display the structure of the table.
6. Write SQL queries to find:
   (a) Students in branch `CS` whose names start with `A`.
   (b) Students whose names contain `AN`.
   (c) Students whose second letter is `A` and whose names end with `N`.
   (d) Students in branch `EC` whose names start with `AL`.
7. Display the students in ascending order of name.
8. Find the total number of students using the `COUNT()` function.

— End of Set 4 —